

Dirb

Cybersecurity Internship – Phase 1, Task 7

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What is DIRB?

DIRB is a scanner for web content. It searches for hidden or already-existing Web objects. It essentially operates by attacking a web server using a dictionary and examining the answer. Although you can use your own custom wordlists, it comes with a collection of preconfigured assault wordlists for convenience. Additionally, DIRB can occasionally be used as a traditional CGI scanner; however, keep in mind that it is a content scanner rather than a vulnerability scanner. The primary goal is to support expert web application audits, particularly in tests pertaining to security. It fills in some gaps left by traditional online vulnerability scanners. DIRB searches for particular web objects that are not detectable by other generic CGI scanners. It doesn't scan for vulnerabilities or potentially susceptible site content.

Tool DIRB is built-in Kali Linux^[24]. Open the terminal and type the following command to get an overview of the tools included in the package:

```
$ dirb
DIRB v2.22
By The Dark Raver

dirb <url_base> [<wordlist_file(s)>] [<options>]

NOTES
=====
<url_base> : Base URL to scan. (Use -resume for session resuming)
<wordlist_file(s)> : List of wordfiles. (wordfile1,wordfile2,wordfile3...)

HOTKEYS
=====
'n' -> Go to next directory.
'q' -> Stop scan. (Saving state for resume)
'r' -> Remaining scan stats.

===== OPTIONS =====
-a <agent_string> : Specify your custom USER_AGENT.
-b : Use path as is.
-c <cookie_string> : Set a cookie for the HTTP request.
-E <certificate> : path to the client certificate.
-f : Fine tuning of NOT_FOUND (404) detection.
-H <header_string> : Add a custom header to the HTTP request.
-i : Use case-insensitive search.
-l : Print "Location" header when found.
-N <nf_code> : Ignore responses with this HTTP code.
-o <output_file> : Save output to disk.
-p <proxy[:port]> : Use this proxy. (Default port is 1080)
-P <proxy_username:proxy_password> : Proxy Authentication.
-r : Don't search recursively.
-R : Interactive recursion. (Asks for each directory)
-S : Silent Mode. Don't show tested words. (For dumb terminals)
-t : Don't force an ending '/' on URLs.
-u <username:password> : HTTP Authentication.
-v : Show also NOT_FOUND pages.
-w : Don't stop on WARNING messages.
-X <extensions> / -x <exts_file> : Append each word with this extensions.
-z <delay> : Add a milliseconds delay to not cause excessive Flood.
```

Default working of Dirb?

In this attack the common.txt is set as a default word list for directory traversal, the protester can use the following command. Open the terminal and type the following command to start the Brussels Directory attack..

- **dirb http://testphp.vulnweb.com/**

Using the common.txt file, the DIRB returns the enumerated directories found within the target URL as shown in the below image.

```
kejav@kejav:~$ dirb http://testphp.vulnweb.com/
Task 3
-----
Task 4
DIRB v2.22
By The Dark Raver
Task 5
-----
START_TIME: Fri Aug 22 14:06:14 2025
URL_BASE: http://testphp.vulnweb.com/
WORDLIST_FILES: /usr/share/dirb/wordlists/common.txt
Task 7
-----
Task 8
GENERATED WORDS: 4612
-----
Scanning URL: http://testphp.vulnweb.com/
-> Testing: http://testphp.vulnweb.com/.htaccess
```

Enumerating Directory with Specific Extension List

There are a lot of situations where we need to extract the directories of a specific extension over the target server, and then we can use the `-X` parameter of the dirb scan. This parameter accepts the file extension name and then searches the given extension files over the target server or machine.

- `dirb http://testphp.vulnweb.com/ -X .php`

```
kejav@kejav:~$ dirb http://testphp.vulnweb.com/ -X .php
Task 2
-----
Task 3
DIRB v2.22
By The Dark Raver
Task 4
-----
START_TIME: Sat Aug 23 19:00:10 2025
URL_BASE: http://testphp.vulnweb.com/
WORDLIST_FILES: /usr/share/dirb/wordlists/common.txt
EXTENSIONS_LIST: (.php) | (.php) [NUM = 1]
Task 7
-----
GENERATED WORDS: 4612
-----
Scanning URL: http://testphp.vulnweb.com/
^C> Testing: http://testphp.vulnweb.com/.cvs.php
```

The above command will extract all directory paths related to php extension as shown in the following image.

Save Output to Disk

For the purpose of the record maintenance, better readability, and future references, we save the output of the dirb scan onto a file. To this, we will use the parameter `-o` of the dirb scan. We can save the output of the dirb scan in a text file.

- `dirb http://testphp.vulnweb.com/ -o commonScan.txt`

The above command will generate a **commonScan.txt** file at the desktop of the enumerated directories.

```
kejav@kejav: ~$ dirb http://testphp.vulnweb.com/ -o commonScan.txt
```

Now that we have successfully executed the command, now let's traverse to the location to ensure whether the output has been saved on the file or not. In this case, our location for output is **/home/kejav/commonScan.txt**.

Not recursively (-r)

The dirb scan, by default, scans the directories recursively. It means it scans a directory and then traverses inside that directory to scan for more subdirectories. But in some scenarios, where time is insufficient, we set the dirb to not scan recursively. This can be achieved using the **-r parameter**.

- **dirb <http://127.0.0.1:42001/index.php> -r**

```
kejav@kejav: ~$ dirb http://testphp.vulnweb.com/
DIRB v2.22
By The Dark Raver
START_TIME: Sat Aug 23 01:43:41 2025
URL_BASE: http://testphp.vulnweb.com/
WORDLIST_FILES: /usr/share/dirb/wordlists/common.txt
OPTION: Not Recursive
GENERATED WORDS: 4612
---- Scanning URL: http://testphp.vulnweb.com/
-> Testing: http://testphp.vulnweb.com/_source
```